

Kidus Yohannes

AI/ML Engineer

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SUMMARY

ML Engineer with 3+ years of experience designing and deploying production-grade machine learning and AI systems on Azure. Deep hands-on experience with the full Azure ML stack, **Azure OpenAI, Azure Machine Learning, Azure Cognitive Search, and Azure DevOps**, alongside end-to-end MLOps practices including model versioning, CI/CD pipelines, Docker, and Kubernetes. Proven track record optimizing ML pipelines for latency, cost, and scalability in real production environments. Proficient in Python, TensorFlow, PyTorch, and Scikit-learn with strong experience integrating ML models into LLM-based systems.

EDUCATION

Wollo University

09/2019 – 10/2024

Bachelors of Science in Software Engineering

Relevant coursework: Data Structures, Algorithms, Software Engineering, Deep Learning

SKILLS

Languages:

React, Next.js, NestJS, Flask, FastAPI, TensorFlow, PyTorch, OpenCV, Express.js

Cloud & Azure

Azure OpenAI, Azure Machine Learning, Azure Cognitive Search, Azure DevOps, Azure Blob Storage, Azure Container Registry, Azure Kubernetes Service (AKS)

AI & Machine Learning

RAG Systems, Semantic Kernel, LLM Integration, Prompt Engineering, Embeddings, Vector Databases, Azure OpenAI, Hugging Face Transformers

Frameworks & Tools:

React, Next.js, NestJS, Flask, TensorFlow, OpenCV

Infrastructure & DevOps

Docker, Kubernetes, CI/CD Pipelines, Redis, Nginx

Data & Search

Azure Cognitive Search, FAISS, Elasticsearch, PostgreSQL, MongoDB

PROFESSIONAL EXPERIENCE

Lead AI/ML Engineer

08/2024 – present

Shega

- **Deployed and managed all ML infrastructure on Azure**, leveraging Azure OpenAI Service, Azure Cognitive Search, and Azure Blob Storage for a production-scale intelligent platform.
- **Architected an advanced RAG-based AI system** using Semantic Kernel and Azure OpenAI, enabling autonomous multi-step reasoning, tool usage, and dynamic decision-making across workflows.
- **Established MLOps practices** including model versioning, automated evaluation pipelines, and CI/CD deployment via Azure DevOps, ensuring reproducibility and reliability across releases.
- **Reduced Azure inference latency by 40% and cut costs** through prompt caching, dynamic token management, and selective context injection.
- **Designed scalable retrieval pipelines** including intelligent chunking, embedding generation, hybrid vector-keyword search, and semantic reranking for high-precision outputs.
- **Built a modular agent framework** supporting planners, skills, memory layers, and tool invocation, enabling extensible and adaptive AI capabilities.
- **Mentored team members and established engineering standards** for scalability, maintainability, and continuous improvement of AI systems.

ML Engineer

06/2025 – 02/2026 | Remote

BrainBite

- **Directed the Azure AI Search migration** from Supabase — configuring indexers, skillsets, and semantic ranking — significantly improving query performance and system reliability.
- **Owned the Azure-based deployment architecture** for AI content pipelines, managing infrastructure across Azure Container Registry and CI/CD automation via Azure DevOps.
- **Optimized end-to-end AI pipelines**, reducing processing time from 87 seconds to 3 seconds, dramatically improving user experience and system efficiency.
- **Designed and implemented advanced vector-based retrieval systems** leveraging multiple indexing strategies to improve semantic understanding and relevance of results.
- **Led a cross-functional engineering team** to deliver AI-driven content generation systems with high scalability and performance requirements.

Chief Technology Officer

06/2024 – 05/2025 | Remote

Alenelachehu Charitable Organization

- **Led the full development lifecycle of a scalable mental health platform**, from system architecture to deployment, increasing service reach by 300%.
- **Designed and implemented backend systems and RESTful APIs** using Node.js and PostgreSQL, ensuring high availability and performance under load.
- **Built and deployed a Python-based Telegram bot** integrated with backend systems, achieving 99.9% uptime and seamless user interaction.

PROJECTS

Amharic Sign Language Translation AI

08/2024 – present

- **Developed a computer vision-based sign language recognition system** using TensorFlow and OpenCV for real-time gesture interpretation.
- **Trained custom deep learning models on annotated datasets**, improving recognition accuracy and robustness across different gesture variations.
- **Designed a real-time processing pipeline** including hand tracking, feature extraction, and classification for practical accessibility use cases.

Health Hub Ethiopia

04/2024 – 06/2024

Teleconsultation Platform

- **Architected and developed a telemedicine platform** integrating video consultations, scheduling, and secure patient interactions.
- **Increased patient engagement by 50% within three months** through improved UX and system reliability.
- **Designed scalable backend systems using NestJS and PostgreSQL**, supporting high concurrency and seamless performance.

Spokenly — Premium AI Transcription Platform

spokenly.kidus-yohannes.engineer (2025)

- **Built a full-stack AI transcription platform** delivering high-accuracy audio-to-text conversion powered by state-of-the-art speech recognition models.
- **Designed and implemented the end-to-end pipeline** covering audio ingestion, preprocessing, model inference, and structured output delivery.
- **Developed a clean, responsive frontend** and RESTful backend API to support real-time and batch transcription workflows.